

Lily Zhang

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RESEARCH INTERESTS

Noncommutative geometry and Poisson geometry, with particular interests in index theory, operator-algebraic methods in geometry and topology, and the geometry of Lie groupoids, Lie algebroids, and foliations.

EDUCATION

Texas A&M University

College Station, TX

Ph.D. in Mathematics

Present

- Advisor: Prof. Guoliang Yu

Washington University in St. Louis

St. Louis, MO

Master of Arts in Mathematics

May 2026

- Advisor: Prof. Xiang Tang
- Thesis: *A Generalization of Molino's Theory to Riemannian Groupoids*

PAPERS IN PROGRESS

- [1] *Explicit representatives for K-theory classes of Roe and localization algebras of Euclidean spaces*, in preparation.
- [2] *A generalization of Molino's theory to Riemannian groupoids*, in preparation.
- [3] *Normal quotients and basic cohomology of blow-ups of proper Lie groupoids*, in preparation.

HONORS & AWARDS

- Charles J. Turck Presidential Honor Scholarship 2019–2023

GRANTS

- JMM Undergraduate Travel Grant Jan 2023
- Macalester College Wallace Scholarly Activities (WSA) Program 2022–2023
- MAA MathFest Student Scholars Travel Grant Aug 2022

CONTRIBUTED TALKS & POSTERS

- Poisson 2026, University of Antwerp & KU Leuven, Belgium Aug 2026 (planned)
- Noncommutative Geometry Seminar, Washington University in St. Louis May 2026
- Szegő Seminar, Washington University in St. Louis Oct 2025
- Joint Mathematics Meetings (JMM), Boston, MA Jan 2023
- MAA MathFest, Philadelphia, PA Aug 2022
- SIMIODE EXPO, Virtual Feb 2022

CONFERENCES & WORKSHOPS ATTENDED

- Groundwork for Operator Algebras Lecture Series, East Lansing, MI *Jun 2026*
- 46th Annual Great Plains Operator Theory Symposium, Iowa City, IA *May 2026*
- Workshop on Poisson Geometry (Gone Fishing), Montréal, QC *Apr 2026*
- Recent Progress in Topological and Geometric Structures in Low Dimensions, Berkeley, CA *Mar 2026*
- Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Berkeley, CA *Jan 2026*
- AMS 2025 Fall Central Sectional Meeting, Saint Louis University, St. Louis, MO *Oct 2025*

TEACHING EXPERIENCE

- Washington University in St. Louis**, Department of Mathematics *St. Louis, MO*
- Grader, MATH 5015 – Introduction to Fourier Series and Integrals *Fall 2025*
 - Grader, MATH 415 – Partial Differential Equations *Fall 2025*
 - Grader, MATH 309 – Matrix Algebra *Spring 2025*
 - Grader, MATH 429 – Proof-Based Linear Algebra *Fall 2024*
- Macalester College**, Department of Mathematics, Statistics & Computer Science *St. Paul, MN*
- TA, MATH 377 – Real Analysis *Fall 2021*
- Ross Mathematics Program** *Hybrid: Columbus, OH & Remote*
- Junior Counselor *Summer 2020*

SERVICE

- Secretary, Association for Women in Mathematics (WUAWM), Washington University in St. Louis *2025–2026*
- Co-organizer, Szegő Seminar, Washington University in St. Louis *2025–2026*
- Speaker, Washington University Math Circle, St. Louis, MO *2024–2025*
- Mentor, MSCS Mentoring Program, Macalester College, St. Paul, MN *2022–2023*